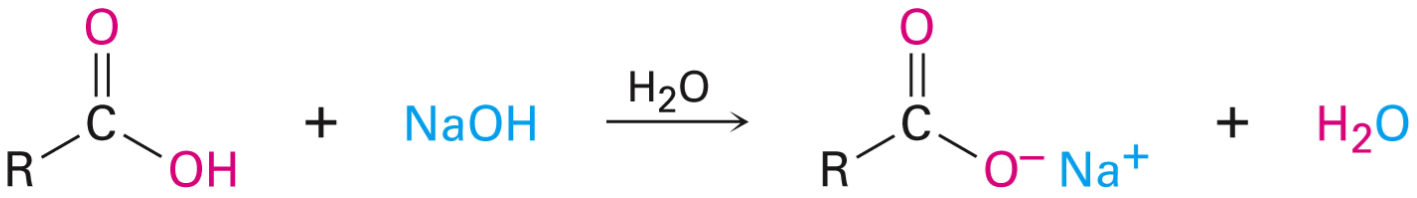
Chapter 20
Carboxylic Acids and Nitriles

Dissociation of Carboxylic Acids

- Carboxylic acids are proton donors toward weak and strong bases
  - Produces metal carboxylate salts, $RCO_2^-\ M^+$
- Carboxylic acids with more than six carbons are slightly soluble in water
  - Conjugate base salts are water-soluble

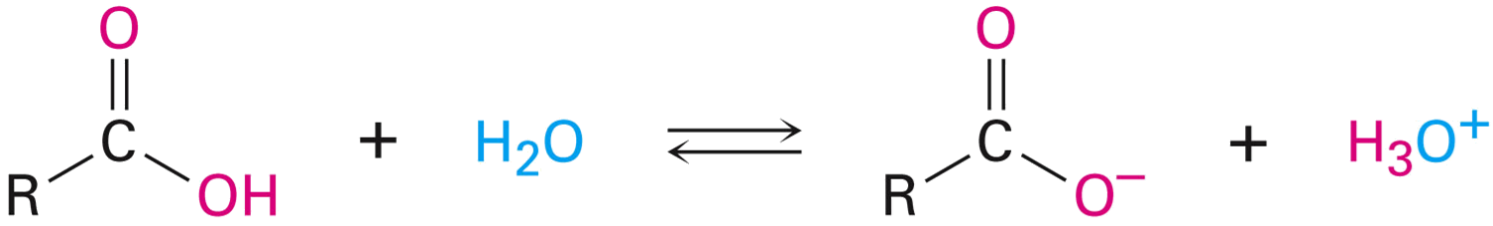


A carboxylic acid (water-insoluble)

A carboxylic acid salt (water-soluble)

Acidity Constant and pKa

- Carboxylic acids transfer a proton to water to give H_3O^+ and carboxylate anions, RCO_2^-
- Acidity constant, Ka, is about 10^{-4} to 10^{-5} for a typical carboxylic acid
  - Gives the extent of acidity dissociation



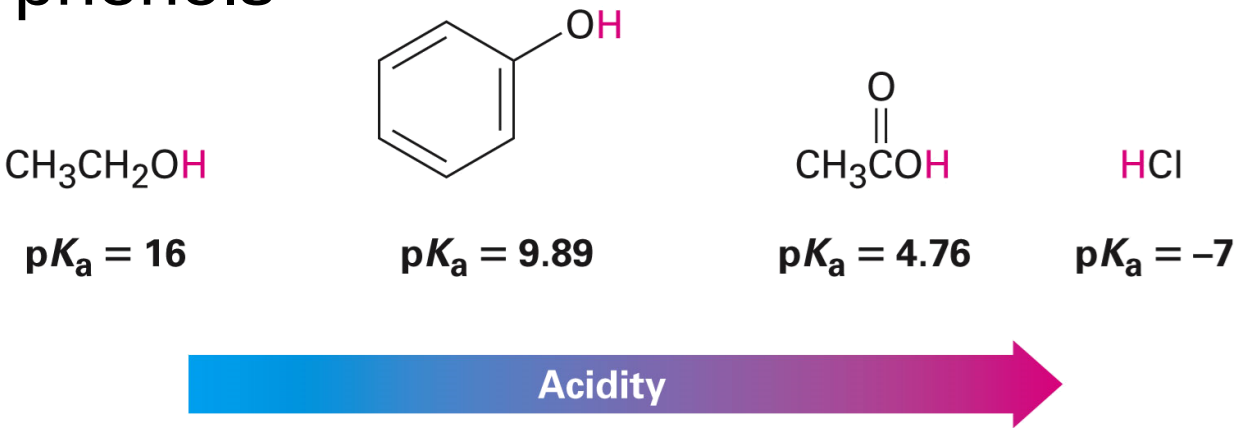
$$K_a = \frac{[RCO_2^-][H_3O^+]}{[RCO_2H]} \text{ and } pK_a = -\log K_a$$

Table 20.3 - Acidity of Some Carboxylic Acids

Structure	K_a	pK_a	
CF_3CO_2H	0.59	0.23	Stronger acid
HCO_2H	1.77×10^{-4}	3.75	
$HOCH_2CO_2H$	1.5×10^{-4}	3.84	
$C_6H_5CO_2H$	6.46×10^{-5}	4.19	
$H_2C{=}CHCO_2H$	5.6×10^{-5}	4.25	
CH_3CO_2H	1.75×10^{-5}	4.76	
$CH_3CH_2CO_2H$	1.34×10^{-5}	4.87	
CH_3CH_2OH (ethanol)	(1×10^{-16})	(16)	Weaker acid

Caboxylic Acid

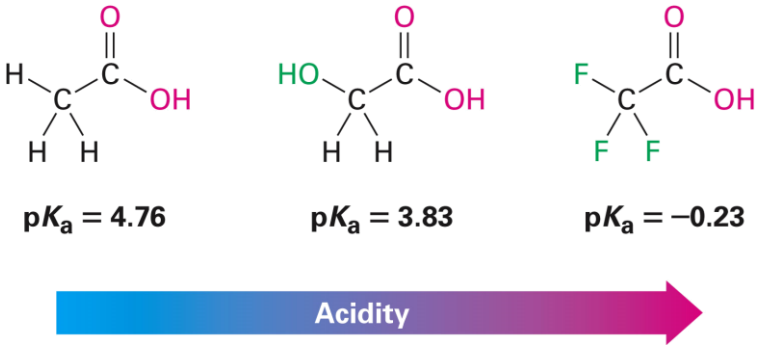
- Carboxylic acids are more acidic than alcohols and phenols



- Carboxylic acid dissociate to give carboxylate ion
  - Carboxylic ion is a stabilized resonance hybrid of two equivalent structures

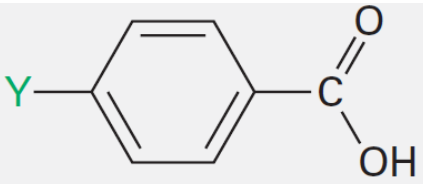
Substituent Effects on Acidity

- Factors that stabilize the carboxylate anion relative to the undissociated carboxylic acid will drive the equilibrium toward increased dissociation and result in increased acidity



- Inductive effects operate through σ bonds and are dependent on distance
  - Substituent moves farther from the carboxyl causing the effect of halogen substitution to decrease

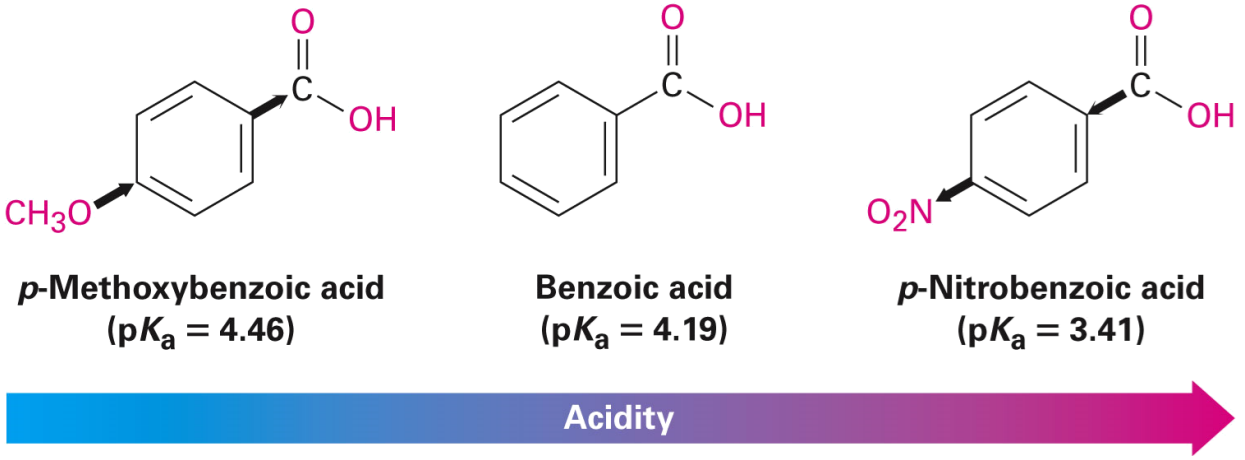
Table 20.4 - Substituent Effects on the Acidity of *p*-Substituted Benzoic Acids



	Y	$K_a \times 10^{-5}$	pK_a	
Stronger acid	$-NO_2$	39	3.41	Deactivating groups
	$-CN$	28	3.55	
	$-CHO$	18	3.75	
	$-Br$	11	3.96	
	$-Cl$	10	4.0	
	$-H$	6.46	4.19	
	$-CH_3$	4.3	4.34	Activating groups
	$-OCH_3$	3.5	4.46	
Weaker acid	$-OH$	3.3	4.48	

Aromatic Substituent Effects

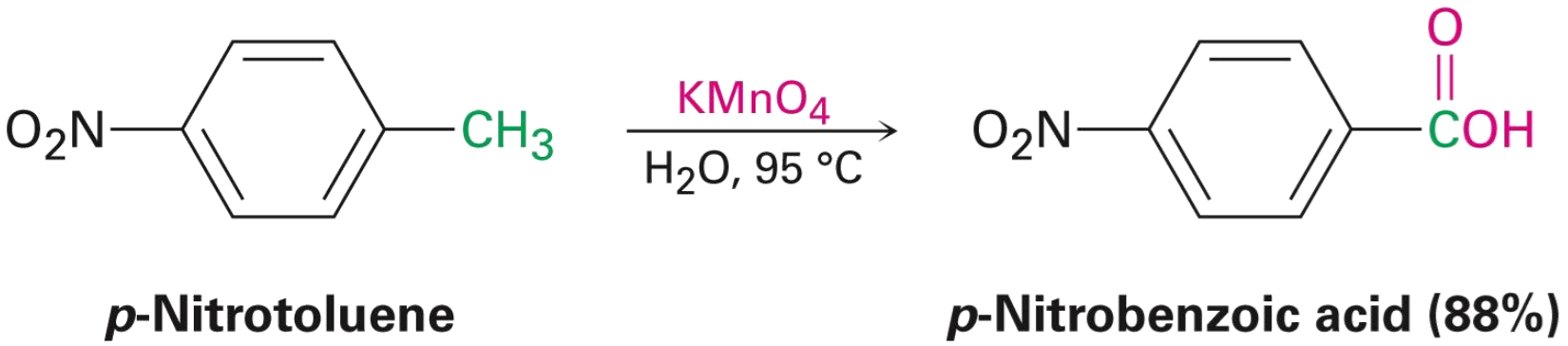
- An electron-withdrawing group increases acidity by stabilizing the carboxylate anion
  - Electron-donating group decreases acidity by destabilizing the carboxylate anion



- By finding the acidity of the corresponding benzoic acid reactivity can be predicted

Preparing Carboxylic Acids

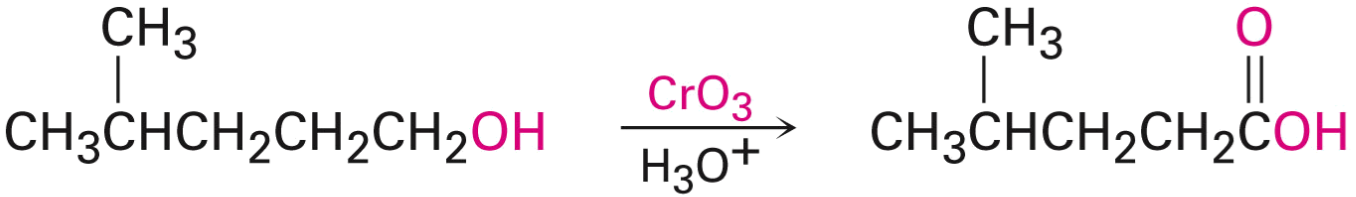
- Oxidation of a substituted alkylbenzene with $KMnO_4$ or $Na_2Cr_2O_7$ gives a substituted benzoic acid
- 1° and 2° alkyl groups can be oxidized
  - Tertiary groups cannot



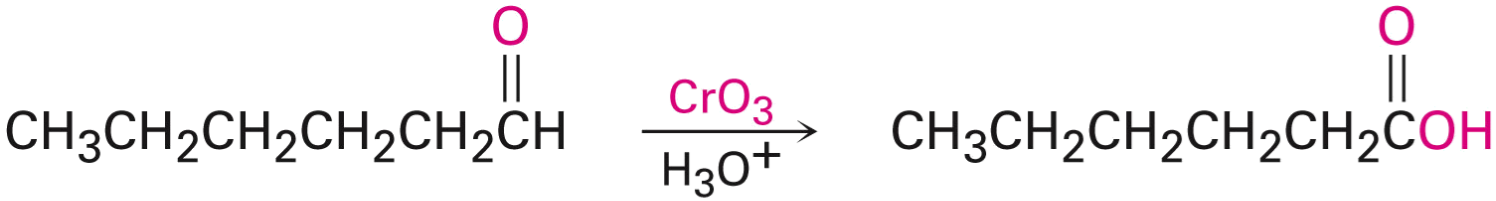
***p*-Nitrotoluene** ***p*-Nitrobenzoic acid (88%)**

Preparing Carboxylic Acids

- Oxidation of a primary alcohol or an aldehyde yields a carboxylic acid
  - 1° alcohols and aldehydes are often oxidized with CrO_3

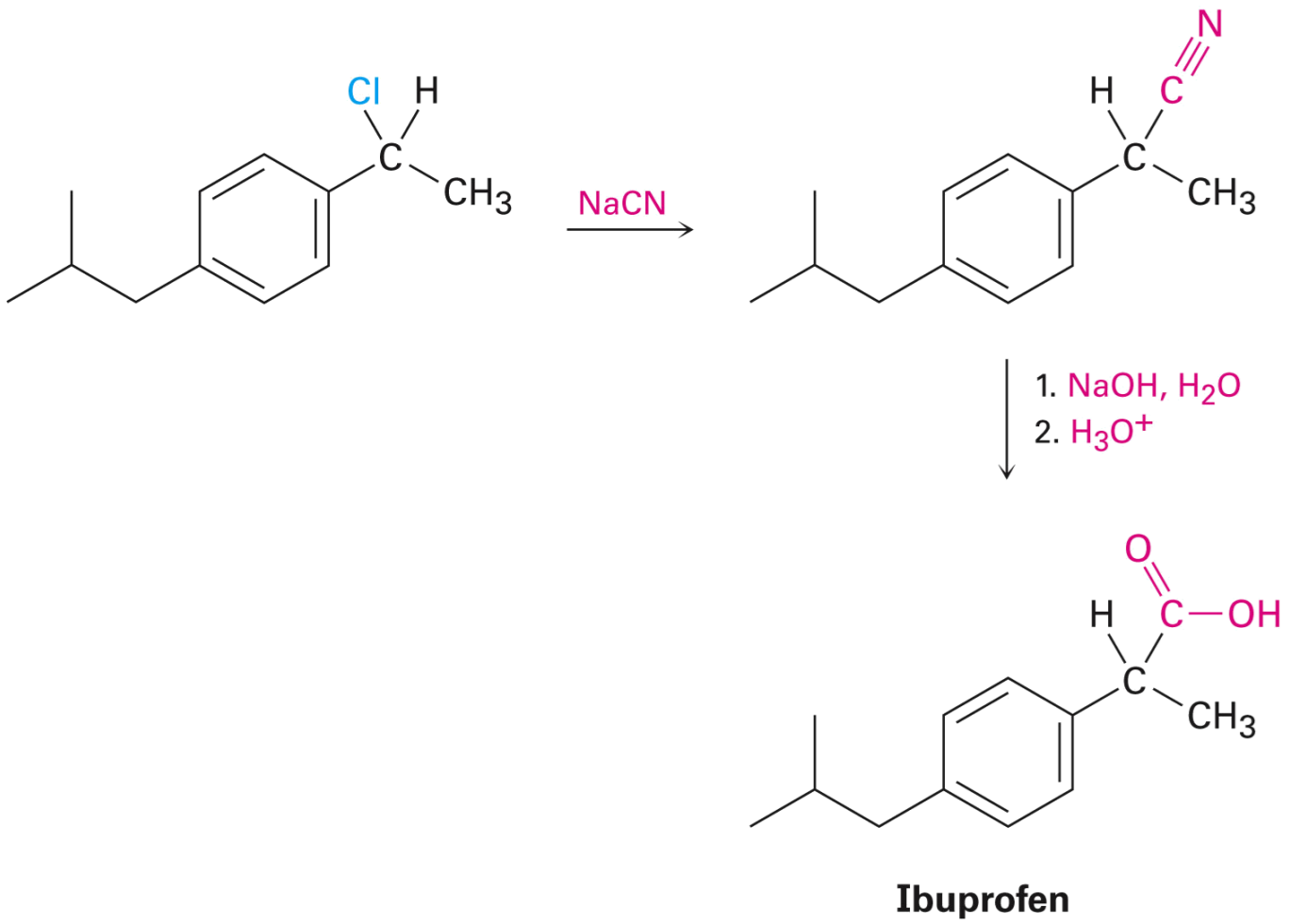


4-Methyl-1-pentanol **4-Methylpentanoic acid**



Hexanal **Hexanoic acid**

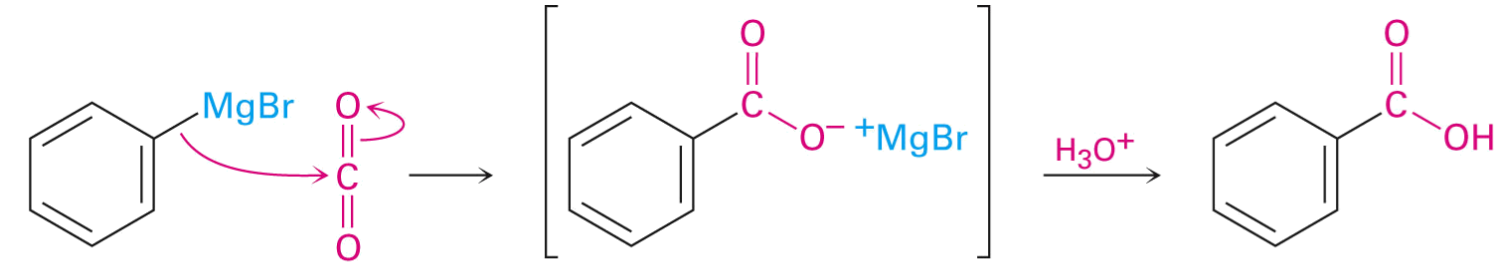
Preparing Carboxylic Acids



Ibuprofen

Preparing Carboxylic Acids

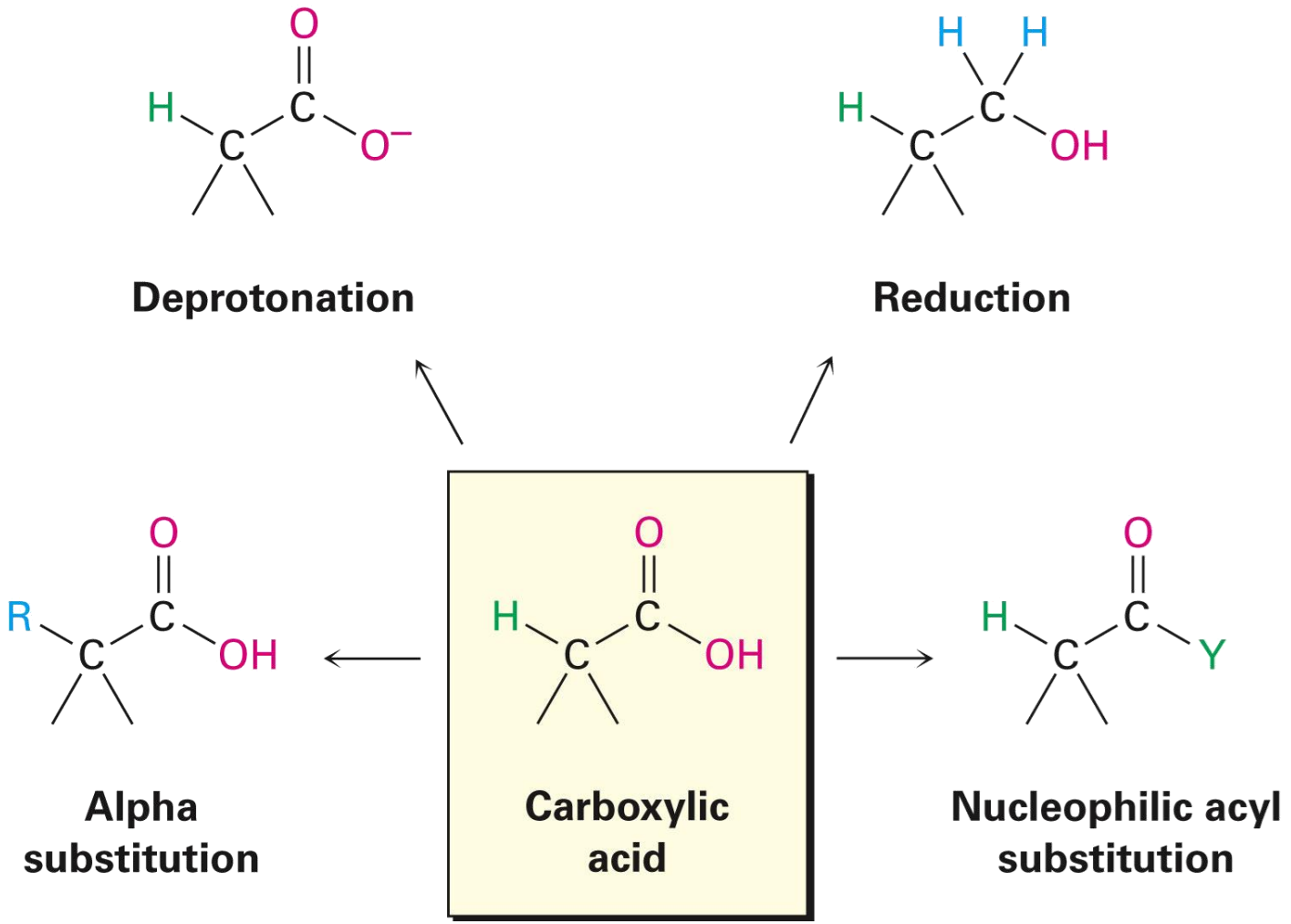
- Carboxylation of Grignard Reagents
  - Grignard reagents react with dry CO_2 to yield a metal carboxylate
  - The organomagnesium halide adds to C=O of carbon dioxide
  - Protonation by addition of aqueous HCl in a separate step gives the free carboxylic acid



Phenylmagnesium bromide

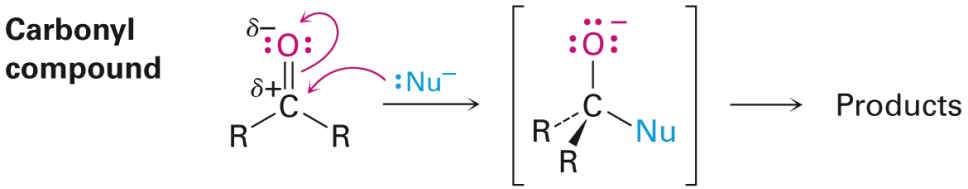
Benzoic acid

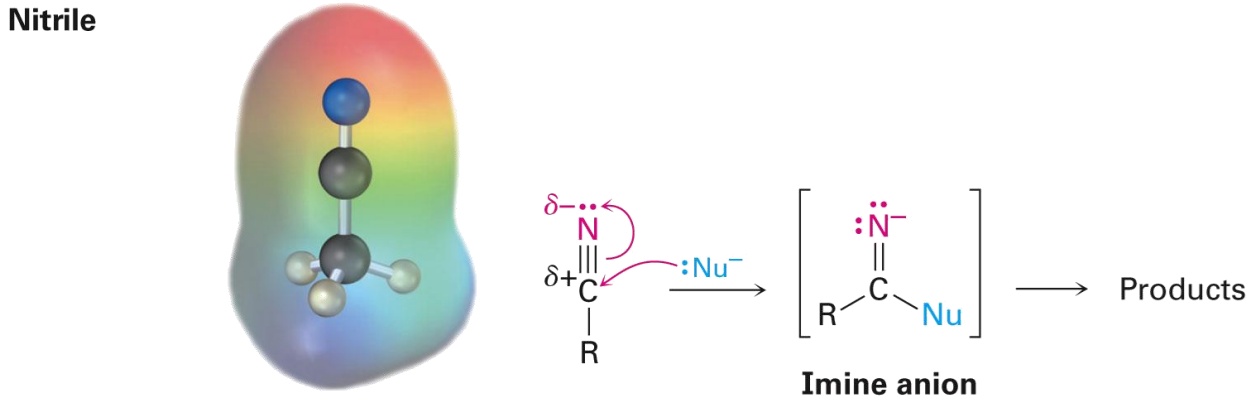
Figure 20.2 - Some General Reactions of Carboxylic Acids



Chemistry of Nitriles

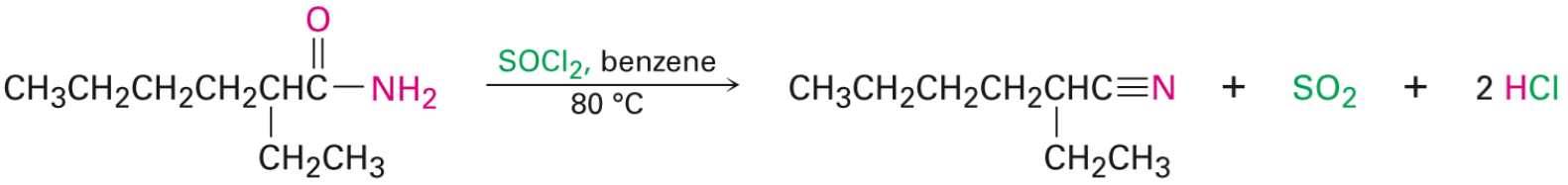
- Reaction of nitriles
  - Strongly polarized and has an electrophilic carbon atom
  - Attacked by nucleophiles to yield sp^2-hybridized imine anions





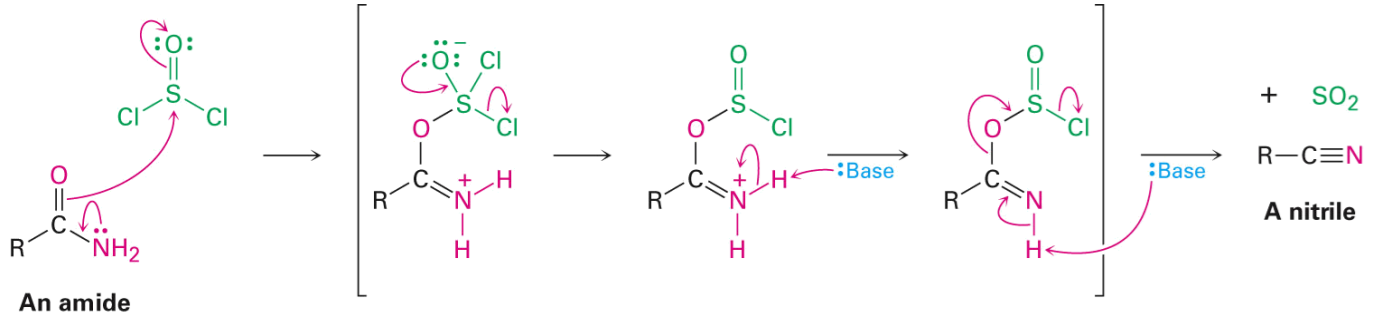
Chemistry of Nitriles

- Preparation of Nitriles
  - S_N2 reaction of CN^- with 1° or 2° alkyl halide
  - Also prepared by dehydration of primary amides $RCONH_2$



2-Ethylhexanamide **2-Ethylhexanenitrile (94%)**

  - Nucleophilic amide oxygen atom attacks $SOCl_2$ followed by deprotonation and elimination



An amide **A nitrile**

Chemistry of Nitriles

- Hydrolysis - Conversion of nitriles into carboxylic acids
  - Nitriles are hydrolyzed in with acid or base catalysis to a carboxylic acid and ammonia

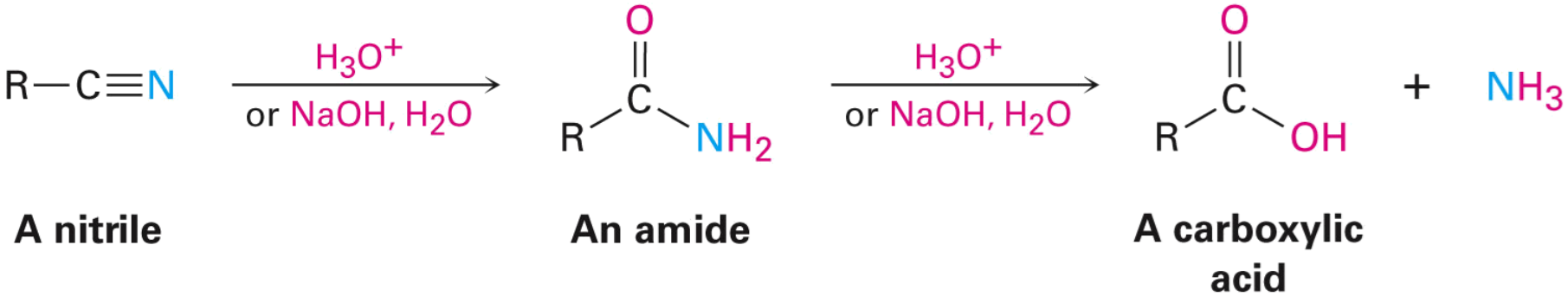
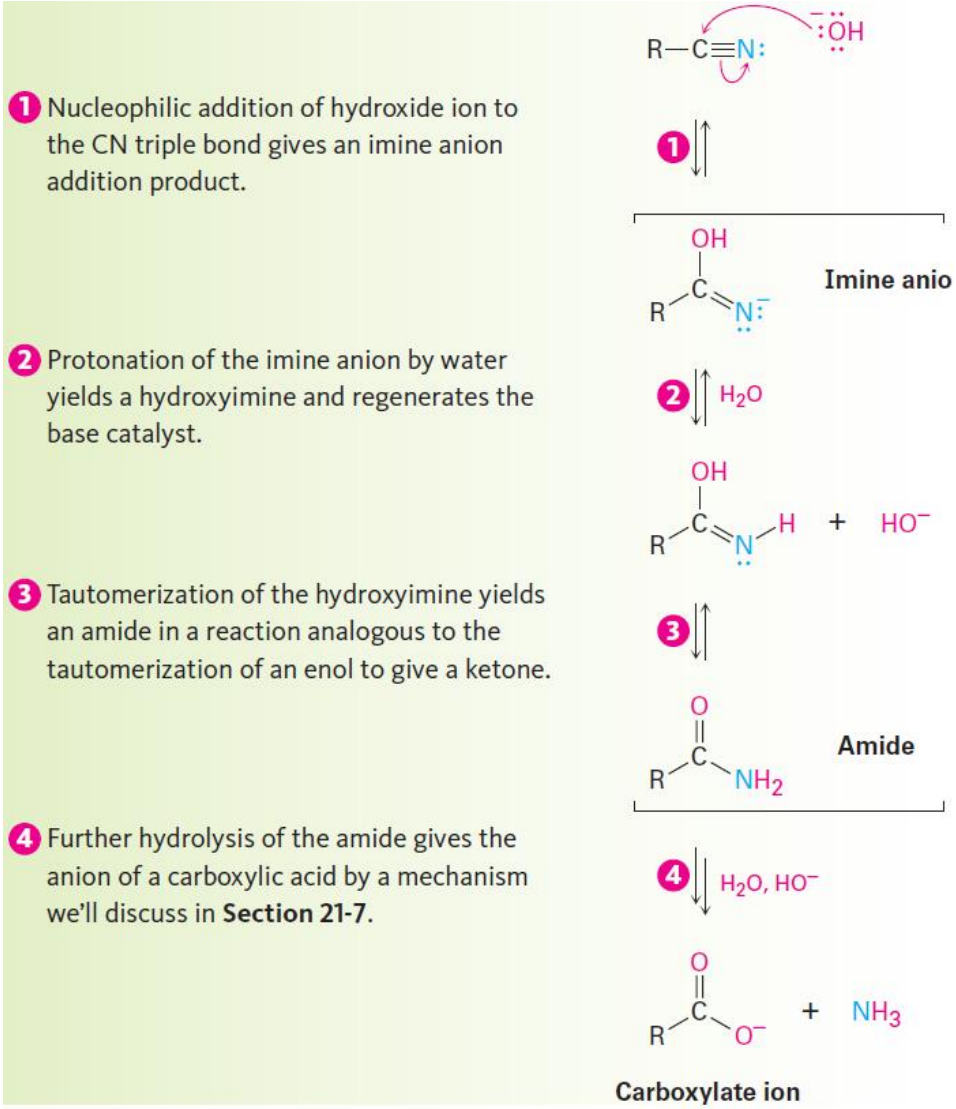
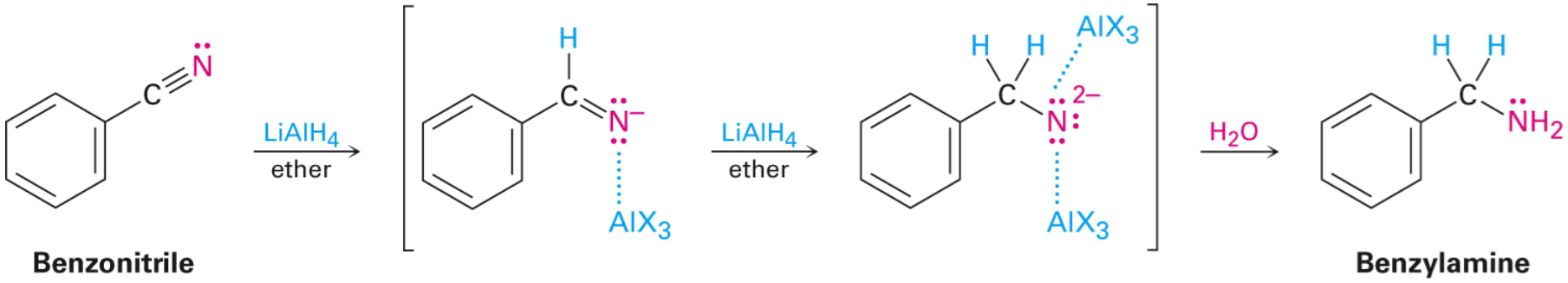


Figure 20.4 - Mechanism of Hydrolysis of Nitriles



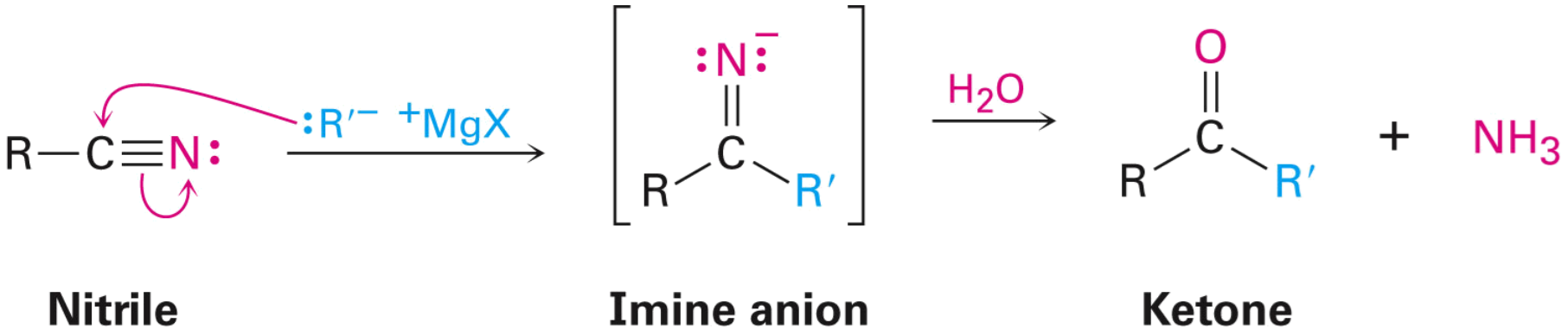
Chemistry of Nitriles

- Reduction - Conversion of nitriles into amines
  - Reduction of a nitrile with $LiAlH_4$ gives a primary amine
  - Nucleophilic addition of hydride ion to the polar C≡N bond, yields an imine anion
  - The C=N bond undergoes a second nucleophilic addition of hydride to give a dianion, which is protonated by water

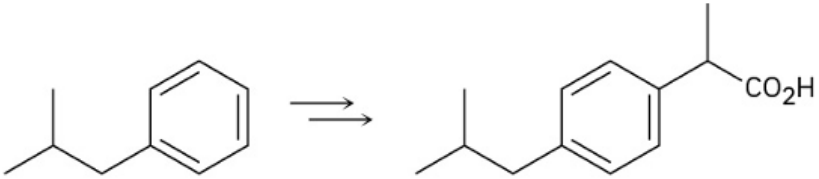


Chemistry of Nitriles

- Reaction of nitriles with Grignard reagents
  - Add to give an intermediate imine anion that is hydrolyzed by addition of water to yield a ketone

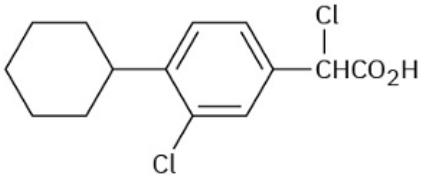


PROBLEM 20-58 Show how you might prepare the anti-inflammatory agent ibuprofen starting from isobutylbenzene. More than one step is needed.



Isobutylbenzene **Ibuprofen**

PROBLEM 20-61 Propose a synthesis of the anti-inflammatory drug fenclorac from phenylcyclohexane.



Fenclorac

PROBLEM 20-65 Identify the missing reagents **a**–**f** in the following scheme:

